

English Glossary: Artificial Intelligence & Machine Learning

Source glossary prepared for SLM-based English-to-Tamil translation demonstration.

Term	Definition
Artificial Intelligence	Technology that enables computers to perform tasks that normally require human intelligence.
Machine Learning	A method in which computer systems learn patterns from data and improve their predictions without being explicitly programmed for every case.
Deep Learning	A branch of machine learning that uses multi-layer neural networks to learn complex patterns from large amounts of data.
Neural Network	A computational model made of connected processing units inspired by the structure of biological neural networks.
Dataset	A structured collection of data used for training, validating, testing, or analysing a model.
Algorithm	A defined sequence of steps or rules used to solve a problem or perform a computation.
Training	The process of adjusting a model's parameters using examples so that it learns patterns from data.
Inference	The process of using a trained model to generate a prediction or response for new input.
Classification	A task in which a model assigns an input to one of several predefined categories.
Natural Language Processing	A field of AI that enables computers to analyse, understand, and generate human language.
Computer Vision	A field of AI that enables computers to interpret and analyse images and videos.
Transformer	A neural-network architecture that uses attention mechanisms to model relationships between tokens in a sequence.
Tokenization	The process of breaking text into smaller units called tokens before it is processed by a language model.
Embedding	A numerical vector representation that captures useful information about a word, token, sentence, or other object.
Prompt	The input instruction or text given to a language model to guide the response it generates.